Esch384 Hash on FPGA: Progress Report

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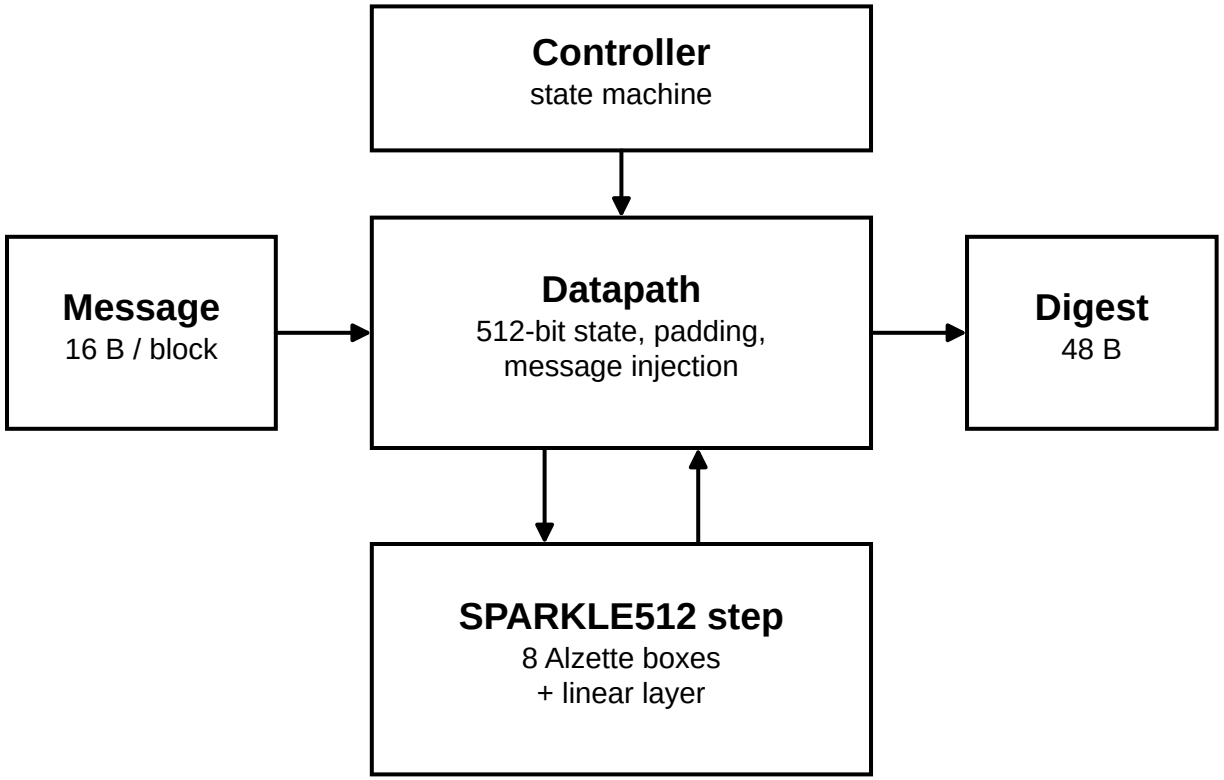
Design: iterative, one step per clock cycle



8 steps per block, 12 on the last block

Target board: Arty A7 (Artix-7), Vivado

Goal: smallest area

Serialize: 8 → 4 → 2 → 1 Alzette boxes

Compare LUTs, flip-flops, clock speed, throughput

Progress

Done: reference model, all 1025 official tests

Done: Alzette, linear layer, SPARKLE512
(all unit tests pass; 2023 LUTs, 521 FFs)

Next: datapath, controller, full hash, synthesis

Risks and challenges

Each block tested alone against official C code

Unclear byte order → checked with C code

Simulator crash → simpler testbench code